Rabiu Onoruoiza Mamman

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Research Interests

Bio-inspired smart materials, artificial muscles, multiscale characterization of engineering materials, soft robotics

Education

The University of Iowa IA, United States
Ph.D. in Mechanical Engineering. CGPA: 4.0/4.0 2022–2026

The University of Iowa IA, United States
M.S. in Mechanical Engineering. CGPA: 4.0/4.0 2022–2025

Federal University of Technology Minna, Nigeria
M.S. in Mechanical Engineering (Industrial and Production) CGPA: 4.53/5.0 2016–2019

Research Experience

Smart Multifunctional Material Systems Laboratory University of Iowa, IA, USA
Graduate Research Assistant (Project 1: Development and Characterization of a Bistable Linear Actuator) 2024–Present

- Designed a vacuum-actuated bistable linear actuator, exploiting snap-through instabilities to achieve faster actuation and passive self-locking.
- Developed parametric CAD and finite-element models capturing negative stiffness and bistability, explaining material-dependent behavior between FDM 3D-printed PLA and compliant resins (Flexible 80A), establishing design rules for stable latching.
- Built a precision pneumatic actuation system and control platform integrating electro-pneumatic regulation, Arduino-based PWM-to-analog control, laser displacement sensing, and NI-DAQ data acquisition for performance evaluation.
- Achieved impulsive snap-through actuation (~4.1 cm stroke in ~0.3 s) while lifting up to 0.5 kg at <65 kPa vacuum pressure, demonstrating peak efficiencies below 50.47%.

Smart Multifunctional Material Systems Laboratory University of Iowa, IA, USA
Graduate Research Assistant (Project 2: Bio-inspired Active Vortex Generators for Underwater Flow Control) 2024–2025

- Developed deployable vortex generators powered by artificial muscles for on-demand flow control for underwater hydrofoils.
- Experimentally demonstrated 25–30% increase in lift coefficient and 8–20% drag reduction at high angles of attack and low Reynolds numbers.
- Implemented real-time electrothermal actuation with PID and L1 adaptive control using feedback from embedded thermistors.
- Integrated MATLAB/Simulink, ROS, and Arduino-based hardware to facilitate real-time control.
- Validated active VG performance through hydrodynamic testing and flow visualization in a recirculating water channel.

Smart Multifunctional Material Systems Laboratory University of Iowa, IA, USA
Graduate Research Assistant (Project 3: Fouling Release Mechanism of an Octopus-Inspired Smart Skin)
2023–2024

- Developed an octopus-inspired, artificial muscle-actuated smart skin that actively releases marine biofilms through controlled out-of-plane surface deformation.
- Developed and experimentally validated a physics-based model linking biofilm critical strain and artificial muscle electrothermal actuation parameters.
- Demonstrated >90% biofilm release efficiency for laboratory-grown biofilms using low voltage (~3.2 V) muscle actuation with real-time PID control.
- Characterized biofilm properties using nanomechanical techniques.
- Quantified biofilm release using a laser confocal profilometer and optical microscopy.

Smart Multifunctional Material Systems Laboratory University of Iowa, IA, USA
Graduate Research Assistant (Project 4: Nanomechanical Properties and Wear Resistance of Palladium Diselenide ($PdSe_2$)) 2024

- Performed nanoindentation and nanoscratch using a Hysitron TI-950 TriboIndenter to quantify hardness, Young's modulus, and creep behavior of $PdSe_2$.
- Conducted strain-rate-dependent nanoindentation and creep testing to extract strain-rate sensitivity relevant to flexible electronics reliability.
- Developed nanomechanical datasets supporting the assessment of ductility, elasticity, and suitability for flexible and wearable electronics applications.

Smart Multifunctional Material Systems Laboratory University of Iowa, IA, USA
Graduate Research Assistant (Project 5: Deployable Vortex Generators for Low-Reynolds-Number Applications Powered by Cephalopod-Inspired Artificial Muscles) 2023–2024

- Developed bio-inspired vortex generators powered by twisted spiral artificial muscles (TSAMs) for adaptive flow control in unmanned aerial vehicles (UMAVs).
- Experimentally validated active VG performance in a wind tunnel at low Reynolds numbers.
- Demonstrated stall delay and lift enhancement compared to clean airfoils and static VG configurations.
- Integrated an electrothermal actuation model with L1 adaptive control to achieve

Mechanical Engineering Laboratory Federal University of Technology, Minna, Nigeria
Graduate Research Assistant (Project 6: Offshore Wind Energy Potential Analysis) 2016–2018

- Conducted offshore wind time-series analysis using probabilistic models (Weibull, Rayleigh, exponential distributions).
- Evaluated wind speed characteristics including and energy potential.
- Matched site-specific wind profiles to commercial turbine models.

Materials Engineering Laboratory Ahmadu Bello University, Nigeria
Undergraduate Final Project 2011–2012

- Developed and characterized a bio-based hybrid polymer composite utilizing polyester reinforced with glass fibers and orange peel particulates.
- Established structure–property relationships improved tensile, flexural, hardness, and fracture properties..

Honors & Awards

- **Winner, Research Open House Poster Presentation**, *University of Iowa* 2025
- **Graduate Student of the Year, Golden Torch Award**, *National Society of Black Engineers* 2025
- **College of Engineering Ambassador Travel Grant**, *University of Iowa* 2024

- **Dean's Excellence Fellowship**, *University of Iowa* 2022
- **Full Graduate School Tuition Scholarship**, *University of Iowa* 2022
- **Best Project Award**, *Association of Commonwealth Universities* 2018
- **Summer School Bursary**, *Association of Commonwealth Universities in collaboration with the Chinese University of Hong Kong* 2018

Publications

[1] **Rabiu Mamman**, Salvatore Garofalo, Luigi Bruno, Leonardo Pagnotta, Giada Risso, and Caterina Lamuta. "Vacuum-Driven Artificial Muscles Exploiting Snap-Through Instability for Fast and Efficient Actuation". Under review, Manuscript ID: SMS-120156. Feb. 2026.

[2] **Mamman, Rabiu Onoruoiza**, Thilina H. Weerakkody, and Caterina Lamuta. "Bioinspired Active Vortex Generators for Underwater Flow Control". In: *Robotics Reports* 3.1 (2025), p. 28350111251365627.

[3] **Mamman, Rabiu Onoruoiza**, Tatum Johnson, Thilina H. Weerakkody, and Caterina Lamuta. "Fouling Release Mechanism of an Octopus-Inspired Smart Skin". In: *Advanced Functional Materials* 34.41 (2024), p. 2406405. DOI: 10.1002/adfm.202406405.

[4] U. Uzun, Parth Kotak, M. A. Shakib, **Mamman, Rabiu Onoruoiza**, S. Daws, C. N. Kuo, and C. S. Lue. "Nanomechanical Properties and Wear Resistance of Palladium Diselenide ($PdSe_2$) for Flexible Electronics". In: *Materials Science and Engineering: B* 304 (2024), p. 117357.

[5] **Mamman, Rabiu Onoruoiza**, Parth Kotak, Thilina H. Weerakkody, Tatum Johnson, Austin Krebill, James Buchholz, and Caterina Lamuta. "Deployable Vortex Generators for Low Reynolds Numbers Applications Powered by Cephalopods Inspired Artificial Muscles". In: *iScience* 26.12 (2023), p. 108369. DOI: 10.1016/j.isci.2023.108369.

[6] **Mamman, Rabiu Onoruoiza**, Oyewole Adedipe, Sunday Albert Lawal, Oluwafemi Ayodeji Olugboji, and Victor Chiagozie Nwachukwu. "Analysis of Offshore Wind Energy Potential for Power Generation in Three Selected Locations in Nigeria". In: *African Journal of Science, Technology, Innovation and Development* 14.3 (2022), pp. 774–789.

[7] Sunday Albert Lawal, Ezekiel Asuku Mayaki, Oyewole Adedipe, Uzoma Gregory Okoro, and **Mamman, Rabiu Onoruoiza**. "Assessment of Three Selected Locations in Nigeria as Offshore Windfarms Using Multi-criteria Decision Procedure". In: *ABUAD Journal of Engineering Research and Development* 3.1 (2020), pp. 90–102.

[8] **Mamman, Rabiu Onoruoiza** and Aliyu Mohammed Ramalan. "Mechanical and Physical Properties of Polyester Reinforced Glass Fibre/Orange Peel Particulate Hybrid Composite". In: *Advanced Journal of Graduate Research* 7.1 (2019), pp. 18–26. DOI: 10.21467/ajgr.7.1.18-26.

[9] N. W. Okafor, Oyewole Adedipe, Fidelis Jonah Usman, J. Y. Jiya, and **Mamman, Rabiu Onoruoiza**. "Production of Biogas from Chicken and Goat Wastes". In: *Nigerian Research Journal of Engineering and Environmental Sciences* (2019).

[10] Oyewole Adedipe, M. S. Abolarin, and **Mamman, Rabiu Onoruoiza**. "A Review of Onshore and Offshore Wind Energy Potential in Nigeria". In: *IOP Conference Series: Materials Science and Engineering* 413.1 (2018), p. 012039. DOI: 10.1088/1757-899X/413/1/012039.

Technical Skills

Prototyping and Fabrication: SLA and FDM 3D printing, resin molding and spin coating, hand layup, thin-walled polymer fabrication

Instrumentation Metrology: Hysitron TriboIndenter (nanoindentation, nanoscratch, and wear testing), TestResources tensile/DMA systems, Keyence LK-G157 displacement sensor, NI-DAQ acquisition system, Keyence laser confocal profilometer, pressure/vacuum regulators, calibration, and troubleshooting
Software Proficiency: CREO Parametric, AutoCAD (transferable to SolidWorks), Abaqus, MATLAB/Simulink, Python, Excel, Power BI, ImageJ, Microsoft Office Suite, Robot Operating System (ROS)

Talks & Presentations

- **Bioinspired Active Vortex Generators for Enhanced Underwater Flow Control**, *Smart Materials, Adaptive Structures, and Intelligent Systems (SMASIS) Conference, St. Louis, MO, United States* 2025
- **Active Vortex Generator for Underwater Flow Control**, *University of Iowa, College of Engineering, Research Open House Poster Presentation* 2025
- **Physics-Based Model for Biofilm Detachment**, *Smart Materials, Adaptive Structures, and Intelligent Systems (SMASIS) Conference, Atlanta, GA, United States* 2024
- **Bio-inspired Vortex Generators for Stall Delay**, *Smart Materials, Adaptive Structures, and Intelligent Systems (SMASIS) Conference, Austin, TX, United States* 2023
- **University of Iowa, College of Engineering, Research Open House Poster Presentation**, *University of Iowa Research Open House* 2023

Teaching

ME:6130 Novel Artificial Muscles and Sensors, University of Iowa, Department of Mechanical Engineering 2023–2025

- Conducted computer lab sessions using MATLAB and Simulink for modeling twisted and coiled/spiraled artificial muscles
- Instructed lab sessions for manufacturing and characterization of novel artificial muscles.
- Grade exams, homework, and proctored exams.

ME:4120 Modern Engineering Materials for Mechanical Design, University of Iowa, Department of Mechanical Engineering

- Developed and delivered senior undergraduate level course on modern engineering materials.
- Taught Finite Element Analysis (FEA) for basic simulation, viscoelastic materials and hyperelastic materials.
- Proctored exams and graded homework.

Service

Laboratory and Safety Manager, *Smart Multifunctional Material Systems Laboratory (SMMs), College of Engineering, University of Iowa* 2024–Present

Grant Review Committee Member, *Graduate and Professional Students Government (GPSG), University of Iowa* 2023–Present

Apex Member, *National Society of Black Engineers, University of Iowa Chapter* 2023–Present